

PAPER_1414_CDF_W_MASS_ANOMALY

Daniel T. Murphy

PAPER_1414 — UQFF: CDF W-mass Anomaly (Bucket J)

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket J

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — formal catalog tuple in Bucket J

Calculator surface: `calculate_higgs_precision({...})`

Closure

``

$\Delta m_W = m_W \times \Lambda \times \beta_i \times \Phi_{\text{res}} / D_{\text{phys}} = 74.24 \text{ MeV} \text{ (2.31\% vs CDF 76 MeV)}$

``

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical primitives. Zero free parameters.

Reference

- Catalog tuple resides in `_higgs_precision_report()` or equivalent surface in `uqff_pure_calculator.py`.

- Related foundational papers: PAPER_646, PAPER_1156, PAPER_1203.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 16, 2026, Youngstown OH.